

GLARUS, Switzerland

WMO: 066850

Lat: 47.0347N

Lon: 9.0669E

Elev: 518

StdP: 95.26

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.8	(c) -8.0	(d) -13.5	(e) 1.2	(f) -8.1	(g) -11.1	(h) 1.5	(i) -7.0	(j) 12.4	(k) 8.5	(l) 10.3	(m) 4.3	(n) 1.0	(o) 140	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	29.7	19.9	27.7	19.0	25.9	18.4	20.8	27.7	19.9	26.2	19.1	24.8	5.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	14.1	24.2	17.6	13.4	22.9	16.9	12.8	21.9	62.4	27.7	59.0	26.3	56.2	24.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.0	7.5	-11.5	32.7	2.3	1.6	-13.2	33.8	-14.5	34.7	-15.8	35.6	-17.5	36.8
			-12.2	22.9	2.4	0.9	-14.0	23.5	-15.4	24.0	-16.7	24.5	-18.5	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	0.1	1.1	5.4	10.2	13.6	17.7	19.0	18.4	14.4	10.5	5.2	1.0		
	DBStd	7.74	4.25	4.23	4.08	3.96	3.81	3.70	3.25	3.26	3.22	3.67	4.06	3.94		
	HDD10.0	1238	308	250	148	46	11	1	0	0	4	39	153	278		
	HDD18.3	3273	567	483	401	245	152	56	31	41	123	242	396	536		
	CDD10.0	1147	0	0	5	51	123	231	280	259	134	56	7	1		
	CDD18.3	141	0	0	0	6	36	53	42	4	0	0	0	0		
	CDH23.3	1118	0	0	0	7	72	312	411	289	25	3	0	0		
	CDH26.7	284	0	0	0	0	9	81	117	76	1	0	0	0		
(14) Wind	WSAvg	2.1	1.6	1.8	2.3	2.9	2.8	2.6	2.3	2.2	1.9	1.7	1.8	1.5		
(15) Precipitation	PrecAvg	1830	109	116	131	125	179	200	216	212	155	121	133	131		
	PrecMax	2366	250	352	262	244	368	303	365	369	289	250	343	322		
	PrecMin	1407	7	36	52	27	48	102	83	50	65	9	0	7		
	PrecStd	206	66	72	63	52	79	57	76	76	54	56	73	72		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.0	18.6	24.2	27.8	31.2	32.4	31.9	25.9	23.5	19.9	17.0		
		MCWB	6.1	6.9	9.6	13.5	16.8	20.6	20.4	20.6	17.8	13.4	10.4	8.6		
	2%	DB	8.6	10.9	16.3	21.7	25.3	29.0	29.8	28.7	23.7	19.7	17.1	9.7		
		MCWB	5.9	5.5	8.9	12.4	15.8	19.8	19.7	20.1	17.3	13.6	9.1	6.6		
	5%	DB	6.7	8.3	14.4	19.9	23.2	27.0	27.5	26.6	21.6	17.5	12.6	7.3		
		MCWB	4.7	4.5	8.2	11.6	14.9	18.6	19.1	19.3	16.4	13.1	9.0	5.6		
	10%	DB	5.2	6.3	12.1	17.7	21.0	24.8	25.6	24.4	19.7	15.8	10.4	5.9		
		MCWB	3.5	3.8	7.2	10.5	14.1	17.8	18.5	18.5	15.6	12.6	8.2	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.0	10.9	14.5	18.5	22.1	22.0	22.0	19.4	15.7	11.2	8.7		
		MCDB	9.8	11.1	15.9	22.5	26.2	29.8	29.4	28.7	24.1	18.6	15.7	14.9		
	2%	WB	6.0	6.3	9.6	13.2	16.9	20.3	20.6	20.7	18.0	14.5	10.2	7.1		
		MCDB	8.0	10.0	14.9	20.5	23.4	27.3	27.6	27.2	22.4	17.6	14.7	9.3		
	5%	WB	4.8	5.0	8.6	12.1	15.6	19.3	19.7	19.6	16.9	13.6	9.4	5.8		
		MCDB	6.7	7.9	13.3	18.6	21.2	25.7	26.1	25.5	20.8	16.7	12.9	7.3		
	10%	WB	3.7	3.8	7.5	11.0	14.5	18.3	18.9	18.8	15.9	12.8	8.3	4.6		
		MCDB	5.2	6.1	11.6	16.6	19.9	23.9	24.7	23.7	19.0	15.4	10.6	6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.4	8.4	9.9	9.8	9.8	9.7	9.0	8.0	7.6	6.1	5.2		
		MCDBR	7.3	9.7	12.4	13.8	14.1	13.7	13.6	12.7	10.9	10.1	9.3	7.2		
		MCWBR	4.8	5.7	6.9	7.0	7.0	6.6	5.9	6.2	5.8	5.6	5.0	4.7		
		MCWBR	7.0	8.7	11.1	12.6	12.3	12.4	12.1	11.5	9.9	8.8	8.1	6.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.279	0.308	0.355	0.401	0.420	0.428	0.428	0.418	0.385	0.352	0.312	0.277		
		taud	2.337	2.268	2.228	2.199	2.216	2.245	2.246	2.289	2.363	2.427	2.450	2.385		
		Ebn at Noon	810	853	860	851	847	840	833	826	823	797	767	770		
		Edh at Noon	73	97	118	134	137	134	132	122	103	83	66	62		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.36	2.26	3.47	4.39	4.89	5.68	5.57	4.85	3.70	2.39	1.40	1.12		
		RadStd	0.16	0.28	0.48	0.67	0.49	0.52	0.50	0.47	0.37	0.27	0.19	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	+0.67	+0.71	N/A	N/A	N/A	N/A		
(47) Regional (122 neighbors)		N/A	N/A	N/A	+0.69	+0.40	+0.32	N/A	N/A	+57	+15		

Nomenclature: See separate page